Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/03/2006
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Analysis of a Novel Sensory Mechanism in Root Phototropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
Organization Name:	University of North Carolina-Greensboro		
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	10/01/2000	**End Date:**	07/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension to 07/31/2008 per A. Chu (5/07) Note anniversary date changed to 8/1; start date remains the same (per A. Chu 4-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hangarter, Roger (Indiana University Bloomington) Edelmann, Richard (Miami University)		
Grant/Contract No.:	NCC2-1200		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of Arabidopsis. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. The phototropic response in roots is much weaker than the graviresponse, which competes with and often masks the phototropic response. It was through the use of mutant plants with a weakened gravitropism that we were able to identify the activity of the red-light-dependent phototropic system. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. In this project, we plan to use the European Modular Cultivation System (EMCS) on the ISS. We are working toward a launch date of July 2006 on STS-121 for the EMCS and our experiments. This information is important for growing plants on long-term space missions as part of life support systems and advances the current exploration agenda at NASA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	Past year's Accomplishments--Research/Development Activities 1. First flight build at ARC. We had a complete flight build of 24 EC's (experimental containers) in July 2005. This build was based on a launch date for STS-121 in September 2005. The post-doc and ARC staff performed the final hardware build under careful QA supervision. The pressure seal on the EUE in EC's was tested in the ERM-2. The PI traveled to ARC to approve of the completion of the EUE hardware build. 2. Biocompatibility test at ARC. Since the launch of STS-121 was postponed, we did some further biocompatibility tests with seeds and EUE at ARC. 3. ASGSB meeting in Reno, Nevada. In November 2005, the PI and his team gave several presentations on the EMCS project and related ground-based research at the 2005 ASGSB meeting. This is an important international meeting for researchers with spaceflight experiments. 4. Increment 13 science symposium at JSC. There was a symposium for all scientific experiments on Increment 13 of the ISS at JSC. The PI gave a presentation and science overview at this symposium. 5. Second flight build at ARC. The launch date of STS-121 was changed to July 2006. Thus, we had a (second) complete flight build of 24 EC's in March-April 2006. The post-doc and ARC staff performed the final hardware build under careful QA supervision. 6. Hardware handover at KSC. Based on the launch date of July 2006, the PI and ARC staff plan to have a handover of the hardware at KSC in mid-June 2006. 7. Throughout the funding year. There were numerous telecons among the PI team at Miami University, the co-I at Indiana University, and the ARC team. Detailed minutes from each telecon are available upon request. Some highlights: • The final version of the EUE has been developed and tested. Numerous spaceflight experiment feasibility test have been performed.(e.g. seed viability, long term storage, etc.) • We have continued microarray experiments on root phototropism on both WT and phy strains of Arabidopsis. • Post-flight procedures, including handling the frozen specimens, have been developed. • Numerous videocamera and programing tests of the EMCS have been performed. • The computer-based system to convert the images from the video tape to digital images has been developed and tested. • We have built up seed stock (with high germination rates) for the spaceflight experiments. At least two generations of seeds will be available of the wild-type and mutant strains of Arabidopsis.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Molas ML,Correll MJ, Kiss JZ. "Gene profiling studies of red-light-induced signal transduction in roots. " American Society of Plant Biologists, Seattle, WA, July 16-20, 2005. American Society of Plant Biologists Annual Meeting, Abstract book, July 2005 , Jul-2005
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Abstracts for Journals and Proceedings	Kuntz A.L., M.L. Molas, M.J. Correll, J.Z. Kiss. "Gene profiling of Arabidopsis shoots and roots in response to red light. " American Society for Gravitational and Space Biology, Reno, NV, November 2005. Gravitational and Space Biology 2005 Nov;19(1):7 , Nov-2005
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Abstracts for Journals and Proceedings	Medina F.J., I. Matía, F. González-Camacho, J.Z. Kiss, G. Gasset, R. Marco. "Seed germination and seedling growth in space produce alterations in cell proliferation and nucleolar activity of Arabidopsis root cells." Science on the European Soyuz Missions to the International Space Station Symposium, sponsored by ELGRA,Toledo, Spain, June 27-30, 2006. Science on the European Soyuz Missions to the International Space Station abstract book, June 2006 , Jun-2006
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Articles in Peer-reviewed Journals	Palmieri M, Kiss JZ. "Disruption of the F-actin cytoskeleton limits statolith movement in Arabidopsis hypocotyls." J Exp Bot. 2005 Sep;56(419):2539-50. PMID: 16061504 , Sep-2005
Articles in Peer-reviewed Journals	Kumar P, Kiss JZ. "Modulation of phototropism by phytochrome E and attenuation of gravitropism by phytochromes B and E in inflorescence stems. " Physiologia Plantarum 127: in press, 2006 , Jun-2006
Articles in Peer-reviewed Journals	Kiss JZ. "Up, down, and all around: how plants sense and respond to environmental stimuli. " Proc Natl Acad Sci U S A. 2006 Jan 24;103(4):829-30. PMID: 16418288 , Jan-2006
Articles in Peer-reviewed Journals	Matía I, González-Camacho F, Marco R, Kiss JZ, Gasset G, Medina F-J. "Nucleolar structure and proliferation activity of Arabidopsis root cells from seedlings germinated on the International Space Station." Advances in Space Research 2005 Nov;36:1244-53. , Nov-2005
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